Correlation vs. Causality

Pathways to Arts & Humanities Summer Scholars Program

Instructor: Tudor Schlanger

Summer 2026

Review: Opportunity Cost

Question

You won a **free ticket** to see an Eric Clapton concert (which has no resale value). Bob Dylan is performing on the same night and is your next-best alternative activity. Tickets to see Dylan cost **$40**. On any given day, you would be willing to pay up to **$50** to see Dylan. Assume there are no other costs of seeing either performer.

Based on this information, what is the **opportunity cost** of seeing Eric Clapton?

- A. $0
- B. $10
- C. $40
- D. $50

Review: Opportunity Cost — Answer

Answer

The correct answer is **B. $10**.

The opportunity cost of seeing Clapton is what you **give up**: the net benefit of seeing Dylan.

You value Dylan at $50 and the ticket costs $40, so the opportunity cost is $50 – $40 = **$10**.

Outline

The Cat Question

Question

Did the cat leave a dent?

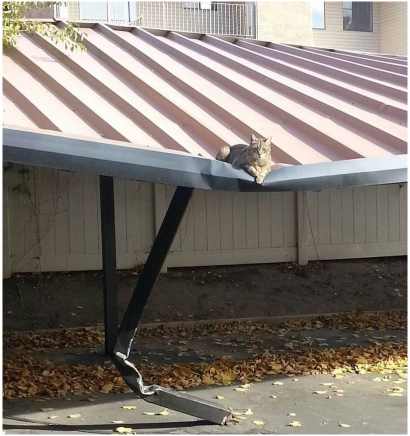


Photo: Internet (fair use, educational)

Catchy Headlines

Question

A 2012 TIME Magazine headline reads:

“Alcohol Does a Body Good? Study Finds It Boosts Bone Health”

Does this mean drinking alcohol strengthens your bones?

Photo: Pexels (free license)

Today's Agenda

By the end of this class, you will be **harder to fool**!

Three things we will practice:

1. **Correlation vs. causality**: how to tell the difference
2. **Misleading graphs**: how data can make you think it's causality!
3. **Chart detective**: how to protect yourself in the future

Outline

1. Can You Trust What You See?

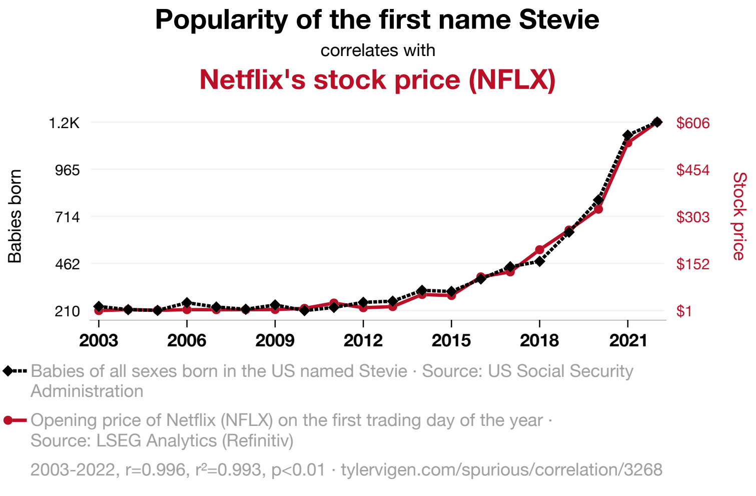
2. Correlation vs. Causality

3. Misleading Graphs and Chart Detective

4. Same Data, Different Stories

5. Recap

A Suspicious Pattern



Source: tylervigen.com — Spurious Correlations.

What's Going On?

Question

Does the popularity of the name "Stevie" **cause** Netflix's stock price to rise?

In your group, come up with:

1. One **funny** explanation (technically possible but absurd)
2. One sentence explaining why the chart **does not prove** causality

Discuss with your group for 1 minute . . .

What's Going On?

Answer

The chart shows a **pattern**, not a cause.

Both the name "Stevie" and Netflix's stock price were rising over the same period—likely for completely unrelated reasons.

Two things trending together does not mean one caused the other. Time itself can create fake-looking relationships.

Correlation

Two things **move together**—when one goes up, the other tends to go up (or down).

It tells us there is a **pattern**, but **not why** the pattern exists.

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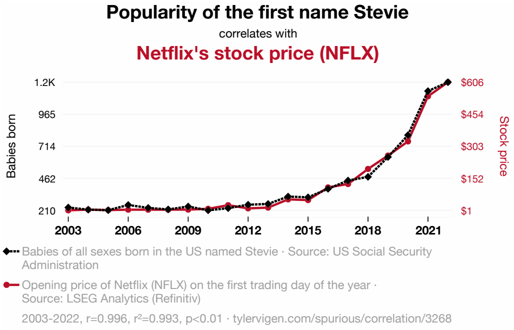


Source: tylervigen.com

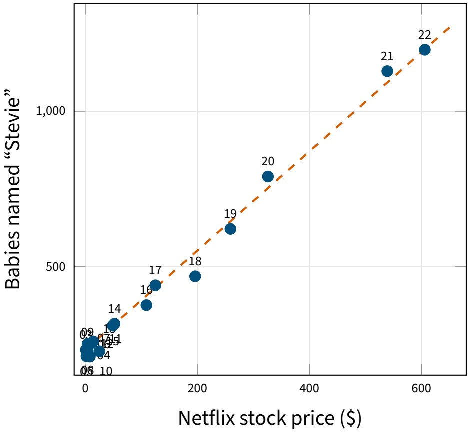
Correlation: Scatter Plot

Correlation

Two things **move together**—when one goes up, the other tends to go up (or down).

It tells us there is a **pattern**, but **not why** the pattern exists.

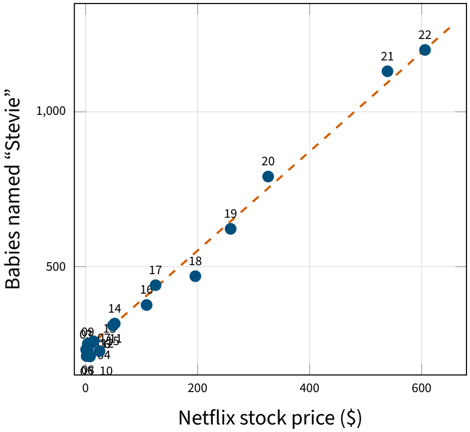
Does Netflix **cause** babies to be named Stevie?



Why Do Correlations Happen?

There are **four** possible explanations when you see two things moving together:

1. **Coincidence**
2. **Causality**
3. **Reverse causality**
4. **Hidden third factor (confounder)**



CAUSALITY

Causality

One thing actually **changes** another.

If you change A, it **affects** B.

Causality

Causality

One thing actually **changes** another.
If you change A, it **affects** B.

Everyday examples:

- Flipping a light switch → light turns on
- Studying more → higher test scores
- Drinking coffee → feeling more alert

In each case, changing one thing **directly affects** the other.

DAGs: A Thinking Tool

A **DAG** (**D**irected **A**cyclic **G**raph) is a map of what causes what.

Drawing the arrows forces you to think about:

- **Direction**—which way does causality flow?
- **What's real**—which connections are causal vs. just correlated?

Economists and scientists use DAGs to reason carefully about cause and effect.

Judea Pearl
Turing Award winner (2011)

Pioneered causal DAGs

Photo: UCLA Newsroom

DAGs: The Simplest Case

> The arrow means: if you **change** A, B changes too.
> The direction matters: A → B is not the same as B → A.

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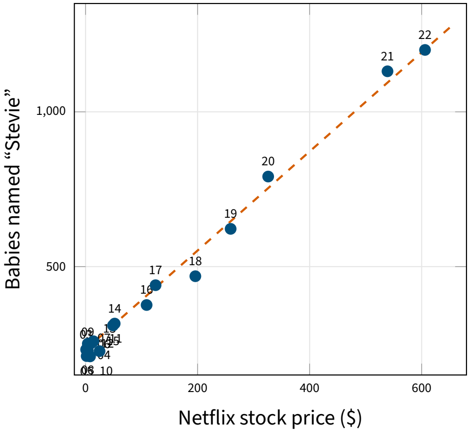

DAGs: Reverse Causality

You *think* A causes B, but the arrow actually runs the other way: **B causes A**. Getting the direction wrong leads to the wrong conclusion.

DAGs: Reverse Causality — Example

DAGs: Reverse Causality — Example

Students who get tutoring have lower grades. Does tutoring **cause** bad grades? No—students who are **already struggling** are the ones who seek tutoring.

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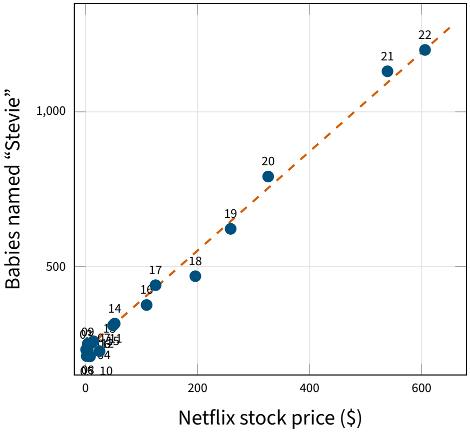
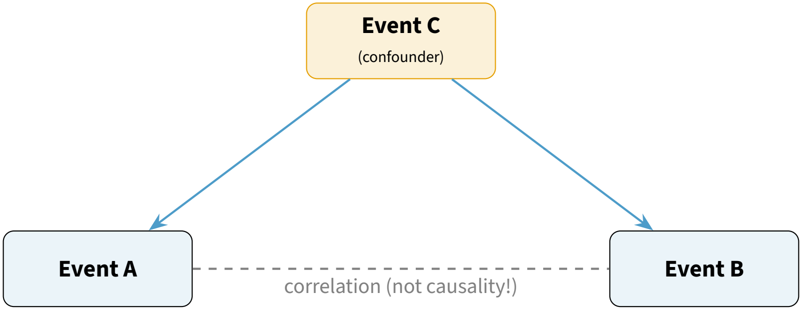
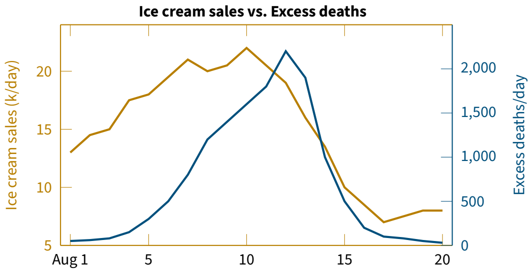


DAGs: The Template



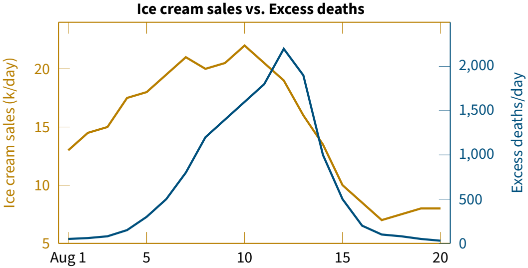
A and B move together—but only because **C drives both**. If you do not see C, you might wrongly conclude that A causes B.

Which Story Is Right?

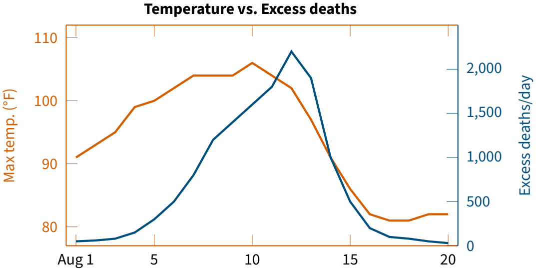


Does ice cream **cause** excess deaths?

Which Story Is Right?

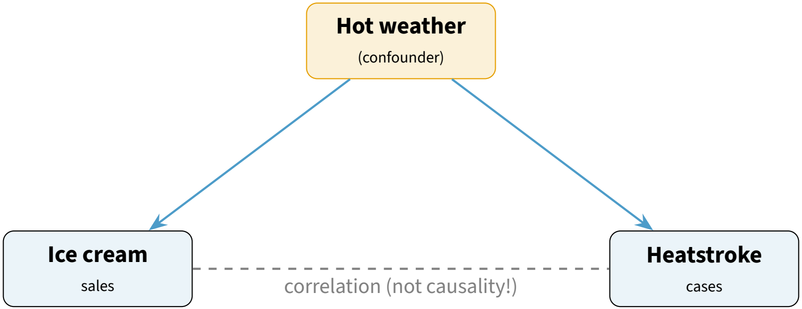


Does ice cream **cause** excess deaths?



No! **Hot weather** drives both.

DAGs: An Example



> Ice cream does not cause heatstroke. **Hot weather** drives both. The DAG makes the hidden driver visible.

From Patterns to Pictures

We have learned that seeing a pattern is **not enough**.

Correlation does not prove causality.

But what if the **numbers are real**—and the **graph itself** is designed to mislead you?

Outline

1. Can You Trust What You See?

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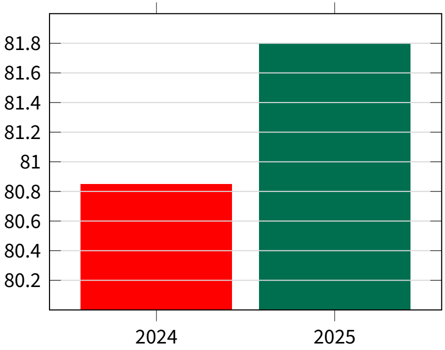
U.S. Steel Production

Question

Look at the steel chart carefully:

1. What are the **units**?
2. Does the y-axis **start at zero**?
3. Compute the difference in **absolute** terms, and then in **percentage** terms.

Total U.S. Steel Production (Mt) 2024 v. 2025



U.S. Steel Production — Answer

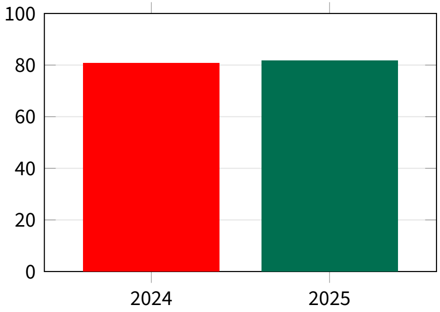
Answer

2024: 80.8 million tons
2025: 81.8 million tons

Increase: 1.0 million tons
That's only ≈1.2%

The bars *look* very different because the y-axis starts at 80, not zero.

Corrected chart (y-axis starts at 0):



Your Turn: Be a Chart Detective

For each chart, use the **Chart Detective** questions with your group:

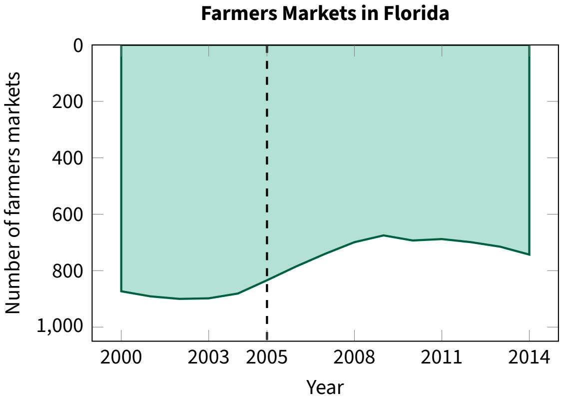
1. What **story** is the title pushing?
2. What are the **units**?
3. Does the y-axis **start at zero**?
4. What **time window** is shown?
5. What **context is missing**?

You have about 1 minute per chart.

Case 1: Farmers Markets in Florida

Chart Detective:

1. What **story** is the title pushing?
2. What are the **units**?
3. Does the y-axis **start at zero**?
4. What **time window** is shown?
5. What **context is missing**?



Case 1: Debrief

Answer

The **y-axis is reversed**—zero is at the **top**, not the bottom. The line going "down" on the page actually means the number is going **up**.

Visual direction ≠ data direction. Our brains see "down" and think "less." This graph exploits that instinct. Always read the axis **before** reading the line.

First fix: Put the y-axis in the normal direction (zero at bottom).

Case 2: COVID New Cases Per Day

Chart Detective:

1. What **story** is the title pushing?
2. What are the **units**?
3. Does the y-axis **start at zero**?
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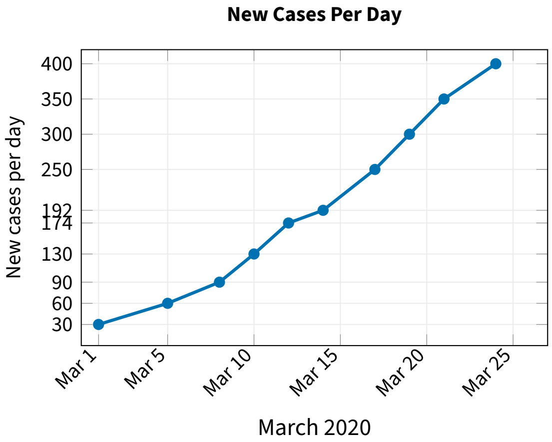
5. What **context is missing**?



Case 2: Debrief

Answer

The y-axis labels are **not evenly spaced**: 30→60 (+30), 60→90 (+30), 90→130 (+40), 130→174 (+44), 174→192 (+18)...

Equal visual distances represent **very different numerical jumps**.

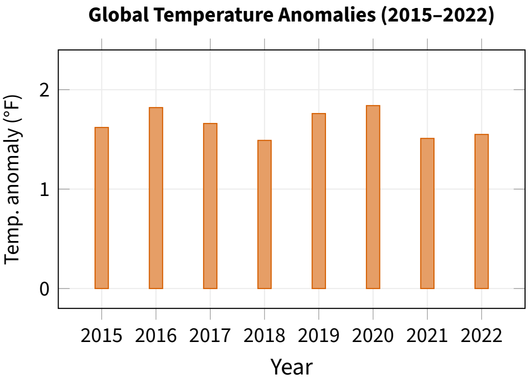
> **If the scale is broken, the story is broken.** The shape of the line stops being trustworthy. The graph can exaggerate some jumps and flatten others.

First fix: Use equal intervals on the y-axis.

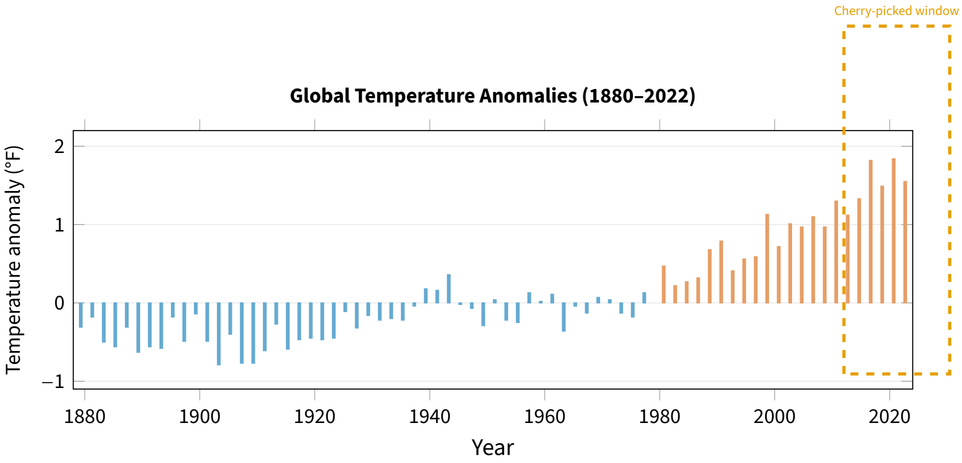
Case 3: Global Temperatures

Chart Detective:

1. What **story** is the title pushing?
2. What are the **units**?
3. Does the y-axis **start at zero**?
4. What **time window** is shown?
5. What **context is missing**?



Case 3: The Bigger Picture



A tiny slice of data can tell a very big lie. The short window hides a clear 140-year warming trend. Always ask: what is the bigger picture?

Three Techniques of Misleading Graphs

Misleading graphs do **not** all work the same way:

Technique	How it misleads	Example
Reversed axis	Visual direction tricks your eye	Florida gun deaths
Broken scale	Equal spaces ≠ equal values	COVID cases chart
Cherry-picked window	Hides the bigger trend	Climate graph

A misleading graph is not one single thing. Sometimes it is visually broken. Sometimes it is technically correct but missing important context. The **Chart Detective** questions catch all of these.

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Same Data, Different Stories

2022 U.S. Inflation Rate (% change, CPI):

Month	%	Month	%
Jan	7.5	Jul	8.5
Feb	7.9	Aug	8.3
Mar	8.5	Sep	8.2
Apr	8.3	Oct	7.7
May	8.6	Nov	7.1
Jun	**9.1**	Dec	6.5

Context: Inflation Reduction Act signed **Aug 16, 2022**.

Activity (see handout):

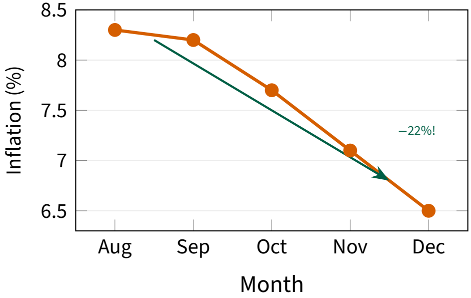
In groups, create **two graphs** from the same data:

- **Graph A:** Tell a "policy worked!" story (misleading but technically true)
- **Graph B:** Tell a more careful, honest story

Also list **3 other reasons** inflation might have fallen (confounders!).

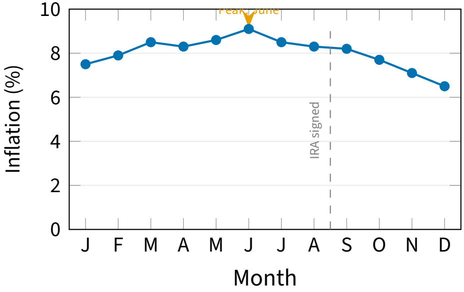
Here Are My Drawings!

Graph A: “Policy Made Inflation Plunge!”



Truncated y-axis, short window, dramatic title

Graph B: “2022 Inflation Rate”



Fair y-axis, full year, inflation peaked before the policy

The Lesson

> **Nobody changed the numbers. What changed was the framing.**
>
> Graph A uses a short window, a truncated axis, and a dramatic title. Graph B shows the full year with a fair scale.
>
> Both use the same data. Only one gives a fair picture.

And remember our earlier lesson: even if inflation *did* fall after the policy, that still does not prove the policy **caused** the decline. What else was changing?

- Energy prices declining
- Supply chains recovering after the pandemic
- Federal Reserve raising interest rates

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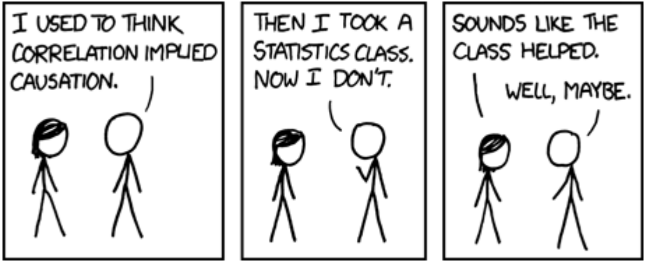
Take-Aways

- **Correlation is not causality.** It can be:
  1. Coincidence
  2. Reverse causality
  3. Confounding factors

- **Even when the numbers are real, the data can mislead you!**

- **Advice:**
  - Slow down
  - Read the axes and the units
  - Ask what is missing
  - Ask what the story is trying to make you believe

And Finally...



Source: xkcd.com/552 — “Correlation” (CC BY-NC 2.5, Randall Munroe)